TRANSFINITTE · HARDWARE TRACK · FINAL TEAM PLAN

HridayMitra

A 75-second daily heart check-in station plus a night-time locket. It warns the family, the ASHA worker and the doctor several days before a heart failure patient needs hospital care. It works in any Indian home: rural or urban, with or without internet, with or without reading skills.

Problem statement: Early Warning of Worsening Heart Failure. This is a monitoring idea, not a medical diagnosis tool.

1. Our idea in one paragraph

Every morning, the patient sits for about 75 seconds on a stool placed on a small weighing platform, with bare feet on four steel plates and one finger in a clip. The station measures **weight, water in the legs (foot impedance), heart rate, rhythm and blood oxygen**. It then asks two questions in the patient's own language, answered with a green or red button. At night, an optional **chest locket** records breathing rate, whether the patient had to sit up to breathe, and restlessness; during the day it counts steps and has an SOS button. All the thinking happens **on the device**, which compares today against the patient's own normal. A **green**, **amber** or **red** light and a voice message show the result. A red result goes out by **SMS and automatic voice call** to the family, the ASHA worker and the doctor, saying *why* it fired. It needs no smartphone, no internet and no reading, and it runs on batteries through power cuts.

2. The three hard rules: how we pass them

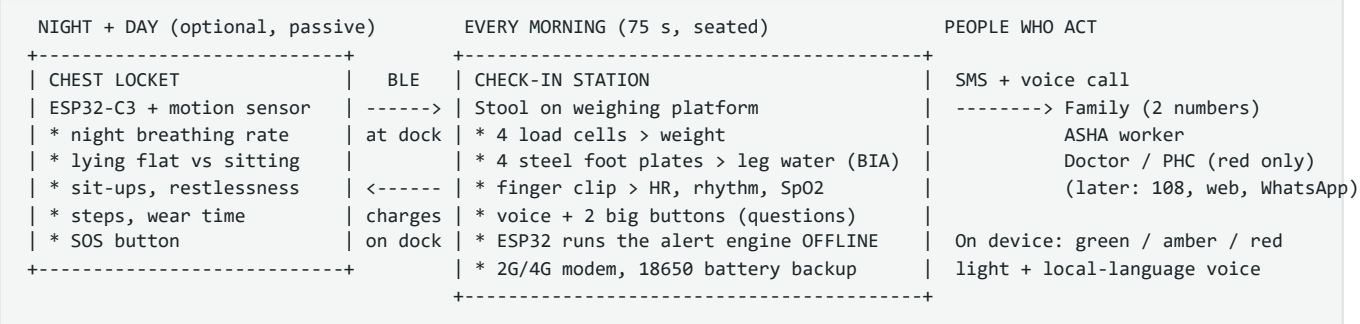
Rule (from the PS)	How HridayMitra meets it	Status
Low effort: nothing, or under 2 min a day; an all-day sticky patch "may not be ideal"	75 s seated check-in + ~10 s to dock and wear the locket = **~85 s/day**. The locket isn't sticky, sits on a cord like normal jewellery, and is **optional**: the station works alone. A short 30 s re-check is only asked on rare amber days	**Pass**
Real home use: little reading, local language, power cuts, weak internet	Voice prompts in the local language, 3 coloured lights, 2 coloured buttons, nothing to read. Battery-first, data stored on the device, decisions made on the device, alerts by SMS and voice call (no data connection needed). Messages wait until the network returns	**Pass**
Low cost: home unit parts under ₹5,000	Station ~₹3,420 + locket ~₹880 = **~₹4,300** (Section 9). About ₹700 margin for price surprises	**Pass, once real quotes are in**

3. Why this approach (what we learned from research)

- **Weight alone is weak.** Daily weight rules catch only **23–35%** of worsening events, with about 4 false alerts per patient per year (FAST trial, SCALE-HF 1).
- **Weight + foot impedance doubles detection.** A scale with foot impedance caught **70% vs 35%**, with fewer alerts (Bodyport SCALE-HF 1; one observational, company-linked study).
- **Night breathing and night heart rate are among the best early signs.** HeartLogic, the best implant system, relies on them. A locket can pick up night breathing cheaply.
- **Bed sensors fail in Indian homes.** Shared beds, charpais, floor mats, blankets and pillows mix up the data, and there's no heart failure evidence for them.
- **Wrist bands get abandoned.** About half of users stop wearing them fully, and wrist pulse readings are noisy whenever the hands move.
- **Programs fail when nobody acts, or people stop using them.** TIM-HF2 worked because a team responded to every alert. Tele-HF and BEAT-HF fell to ~55% use. So we keep the effort tiny and send alerts to people who act (family + ASHA worker).

- **Almost half of wrong alerts come from bad measurements** (TwoCan study). So we check reading quality and ask for a re-check before raising the alarm.

4. System overview



5. What we measure, and for how long

Station: once every morning (signals recorded at the same time)

#	Parameter	Sensor	Duration	PS warning sign
1	Weight	4 load cells + HX711	Best still 5–10 s	Weight gain
2	Foot impedance (water in legs)	4 steel plates, tetrapolar, ~50 kHz tiny current	3 × 10 s + reference check	Fluid in legs
3	Resting heart rate	Finger PPG (MAX30102)	60 s, use last 40 s	Faster resting HR
4	Rhythm irregularity	Same PPG	Same 60 s	Avoids AF false alarms
5	Blood oxygen (SpO_2)	Same PPG	Median of last 20 s	Fluid in lungs (indirect)
6	Breathless when bending (bendopnea)	Voice question + button	~8 s	Breathing trouble
7	More pillows / sat up last night?	Voice question + button	~8 s	Trouble lying flat

Locket: automatic, no effort

#	Parameter	Method	When	PS warning sign
8	Night breathing rate	Chest rise and fall (0.1–0.7 Hz)	60 s every 5 min, 12–5 am, only while still	Faster breathing
9	Minutes propped up / sitting at night	Chest tilt vs gravity	Every 1 s, all night	Trouble lying flat
10	Sit-up episodes, restlessness	Posture change >5 min; movement counts	All night	Breathlessness at night
11	Steps, active minutes	Step counter built into the sensor	6 am–10 pm	Less activity
12	Wear time, SOS	Motion/tilt; button	All day	Reliability, emergencies

Stretch goal: night heart rate from the chest vibration of each heartbeat (seismocardiography), taken only in very still windows.

6. The patient's day (user journey)

When	Patient	System
Night	Sleeps wearing the locket	Records breathing, posture, sit-ups, restlessness

Morning, bath	Puts the locket on the station's dock	Locket charges and sends last night's summary
After toilet, before breakfast and tablets	Sits on the stool, bare feet on plates, finger in clip	Voice: "Namaste Ramesh-ji…" then 60 s of measurement and 2 questions
+75 s	Listens to the result	**Green** "All fine today" · **Amber** "Please check again this evening" · **Red** "We are calling Sunita and ASHA didi"
Day	Wears the locket and lives normally	Counts steps; SOS ready
Weekly	Nothing	Family gets "Device OK, Appa checked in 7/7 days"

Why patients will keep using it for months: it takes under 90 s, reassures them every day ("All fine today"), lets the family see they're being looked after, and the ASHA worker follows up. Family members can weigh themselves on it too; it says "Guest reading" and doesn't count it.

7. How the alert is decided (software)

1. **Quality check:** reject a reading if contact is bad, the patient isn't still, the finger signal is noisy, or it's the wrong person (see identity in Section 8). Retry with a voice prompt.
2. **Personal baseline:** for each signal, a rolling median and spread (MAD) over days −35 to −7. In the first weeks, start from the hospital "dry weight" and population averages.
3. **Worsening scores:** a robust z-score per signal, counting only the worsening direction (weight ↑, impedance ↓, HR ↑, breathing ↑, SpO_2 ↓, activity ↓, upright minutes ↑). Plus published trend detectors: weight moving-average crossover (9 vs 20 days > 0.62 kg) and an impedance cumulative-sum detector (CUSUM) (Gyllensten 2016).
4. **Combined index** across three families: *fluid* (weight, impedance), *heart and lungs* (HR, breathing, SpO_2, night posture), and *symptoms*.
5. **Light colour:**
   - **Amber** = index high once → ask for a 30 s re-check.
   - **Red** = high on 2 of 3 days **and** at least 2 families involved, or one family high for 3 days.
   - The alert clears only when the index drops well below the trigger level (HeartLogic-style 16 → 6), so it doesn't flicker on and off.
6. **Immediate red flags** (no waiting): SpO_2 < 90%, HR > 110 or < 40, new irregular rhythm, "breathless at rest" confirmed twice, or SOS.
7. **Cause tag + explanation:** heart-failure-like / infection-like / AF-like / salty-meal-like / after medicine change / dehydration-like. Example: "*Weight up 1.8 kg in 6 days. Leg water up. Breathing faster at night. Call ASHA.*"

8. Edge cases and how we handle them

Edge case	Our answer
Someone else uses the station	The reading must match the patient's weight band and impedance signature. Otherwise the voice asks "Is this Ramesh-ji?" and logs it as a guest. Guest readings never touch the baseline
Bed partner, blankets, extra pillows	No bed sensor. The locket is on the patient's own body, and the pillow question asks about a *change* from usual
Problem during the day	SOS on the locket; an extra check-in any time runs the red-flag rules at once
Missed check-in	Reminder; SMS to family by 11 am; repeated misses count as a risk sign. Silence never means "all fine"
Locket forgotten or flat	Wear under 12 h counts as "no data", not normal. Voice reminder at the station. The station keeps working without it
Power cut	18650 batteries, device sleeps between uses, data stored on the device, messages queued until power and network return
No internet / weak signal	All decisions on the device; SMS and voice calls need no internet
Can't read	Voice in the local language, coloured lights, colour- and shape-coded buttons
Wet feet, slippers, dust	Tetrapolar measurement cancels contact resistance; a contact check prompts "please remove slippers / dry your feet"
Salty meal, festival food	One-day weight jumps that don't persist are ignored; red needs persistence and 2 signal families
Fever or infection	"Infection-like" tag (sudden HR and breathing rise with flat weight); a different message, still told to the family
Atrial fibrillation	Irregular-rhythm detection → separate "rhythm check" message; HR baseline frozen
Medicine change	Family or ASHA marks it (SMS keyword or button combination) → baseline resets for about 7 days
Device fault	Self-test each check-in (reference resistor, load-cell zero, battery); weekly "device OK" SMS; fault SMS

Pacemaker / ICD	Impedance can be switched off at setup by the doctor (safety not yet verified)
Frail patient, fall risk	Seated use; handle and stool are on the platform, so leaning doesn't spoil the weight

9. Notifications and display

Level	Patient (device)	Family	ASHA	Doctor
Green	Green light + "All fine"	Weekly "device OK" SMS	—	Monthly summary (urban dashboard)
Amber	Amber light + "check again this evening"	Gentle SMS	—	—
Red	Red light + spoken reason	**Automatic voice call** in the local language + SMS	SMS with name and reason	SMS; eSanjeevani link
Urgent	Flashing red + "Call 108 now"	Repeated calls to 2 numbers	SMS	SMS
Fault	"Device needs help"	SMS	—	—

Rural: SMS + voice call, ASHA in the loop. **Urban:** the same, plus Wi-Fi sync to a web dashboard and WhatsApp for children living elsewhere. Indian-language SMS must fit 70 characters per message.

10. Hardware and cost

Custom part (for the schematic and simulation marks)

Tetrapolar foot-impedance front end. The classic ESP32's DAC generates a sine wave, which goes through a band-pass filter into an op-amp current source (Howland type, with current limit and DC blocking). The current is injected through the 2 outer foot plates; an instrumentation amplifier senses the voltage on the 2 inner plates; a precision rectifier and low-pass filter feed the ESP32 ADC. A reference resistor is measured at every check-in. The proof is an LTspice simulation showing that sweeping the skin contact resistance (wet vs dry feet) does not change the reading. This replaces an AD5933 module, which costs about ₹3,500.

Parts list (₹, single-unit retail; estimates are marked and must be replaced with real quotes)

Part	₹	Status
ESP32 classic DevKit (has DAC)	450	Estimate
4 × 50 kg load cells + HX711	544	Verified (₹136/set)
Platform board, stool, non-slip mat, enclosure	450	Estimate
4 stainless-steel foot plates	150	Estimate
Impedance front end (op-amps, in-amp, passives, reference resistor)	250	Estimate
MAX30102 finger clip	150	Weak listing
MAX98357A amp + speaker	330	Amp verified
2 big buttons + 3 LEDs	150	Estimate
2 × 18650 + protection/charger + boost	400	Estimate
Wires, perfboard, locket dock contacts	250	Estimate
SIM800L 2G modem (Airtel/Vi/BSNL SIM)	300	Verified range 150–448
Station subtotal	**3,424**	
Locket: ESP32-C3 SuperMini	350	Estimate
Locket: motion sensor (LIS3DH / BMI270)	150	Estimate
Locket: 150–200 mAh LiPo + charger chip	200	Estimate

Locket: button, vibration motor, case, cord, contacts	180	Estimate
Locket subtotal	**880**	
Home unit total	**~4,304**	**Under ₹5,000, ~₹700 margin**

Upgrades (pick at most one if quotes allow): AD8232 ECG handle for stronger rhythm checking (+~₹650 → ~₹4,950, too tight), *or* a 4G A7670C module so it works on Jio too (bare module +~₹500 → ~₹4,800). The ready-made A7670C board (₹1,900) breaks the budget. **Hackathon build: the base version above.** Show 4G as the product version.

11. Validation plan (software and honest testing)

- **Synthetic cohort:** 200+ virtual patients (the PS asks for at least 20), each with a 30-day warm-up and a 30-day test window. Worsening starts **3–10 days before the hospital visit**, with:
  - weight +1–4.5 kg
  - breathing +1.5–2 per minute
  - heart rate +3–8
  - impedance −5 to −15%
  - steps −15 to −30%
  - more upright minutes at night

  30–40% of episodes have **no weight gain**. Built-in confounders: infection, AF, salty meals, medicine changes, guest users, wet feet, missed days, a forgotten locket.
- **Real data:** test the heart-rate and rhythm code on the PhysioNet **BIDMC CHF** ECG database. Waveforms are simulated with NeuroKit2.
- **Metrics:** sensitivity, specificity, **median lead time**, **false alerts per patient per month**, plus time in alert and PPV. Split by patient (70/30, tuning only on the training patients), with bootstrap confidence intervals.
- **Targets:** beat the weight rule (23–35% sensitivity, ~0.35 false alerts/patient-month). Aim for ≥60% sensitivity at ≤0.21 false alerts/patient-month.
- **Ablations:** weight only → + impedance → + vitals → + locket → + symptoms → + re-check → + identity check.
- **Honesty:** synthetic results prove the logic finds the patterns we built in, not that it's clinically accurate. We'll say so on the results slide.

12. Why we stand out

Judging criterion (weight)	Our edge
Creativity of sensing (20)	Seated "closed-loop" station; DIY tetrapolar foot impedance; bendopnea built into removing slippers; locket that detects sitting up to breathe at night
Ease and long-term use (20)	~85 s/day; voice-guided; guest mode for family; daily reassurance; the station works alone if the locket is forgotten
Software and analytics (20)	12-signal fusion, personal baseline, published trend detectors, hysteresis, re-check step, cause tags, plain-language reasons
Hardware and simulation (15)	Custom impedance circuit with an LTspice proof; Wokwi firmware state machine; full schematic
Science and honest testing (15)	Benchmarked against published weight rules; patient-level split; confounders modelled; limits stated openly
Cost, India fit, pitch (10)	~₹4,300; offline SMS and voice; ASHA + eSanjeevani loop; vs Dozee: "built for the charpai home, not the hospital bed"

13. Team split (5 people, working in parallel)

Person	Owns	Delivers
P1 Hardware	Station + locket design	Block diagram, KiCad schematic of the impedance front end + ESP32, LTspice proof, Wokwi state machine, parts list with links and real quotes
P2 Signals	Raw → features	PPG heart rate, SpO_2, rhythm; locket breathing, posture, steps; BIDMC CHF check
P3 Data	Synthetic patients	200+ patients with confounders, an answer key, and DATA.md explaining how they were made
P4 Engine	Alerts + evaluation	Baseline, scores, light colours, red flags, cause tags, SMS text, metrics, ablations
P5 Product	People-facing side	Voice prompts, family/doctor dashboard, user journey, 2-page report, pitch, GitHub

Fix the shared data formats in the first 30 minutes; everyone builds against sample files and joins up only at the end.

14. Risks and honest limits

- About a third of part prices are estimates, so get quotes first: cost is a pass/fail rule.
- The DIY impedance circuit is unvalidated against real people; Bodyport's evidence is a single company-linked study.
- Lung fluid is detected only indirectly (SpO_2, breathing, symptoms).
- Rhythm checking from a finger clip is weaker than ECG.
- Pacemaker/ICD safety with injected current is not verified; an impedance-off mode is provided.
- SIM800L 2G doesn't work on Jio; the 4G module costs more.
- Results are synthetic; adherence and the response loop in Indian homes are untested.
- It can't predict sudden cardiac death. It is **not an emergency device**.

15. After the hackathon (startup path)

1. Pilot with one district hospital's cardiology team and its ASHA workers, with ethics approval, starting in "shadow mode" (alerts go only to the study team).
2. Customers: children in cities buying for parents (buy or rent), hospitals (90-day post-discharge kits), and government NCD programmes.
3. CDSCO registration (likely Class B/C), IEC 60601 safety testing, and a custom PCB to bring cost down.

Disclaimer: HridayMitra is a monitoring idea, not a medical diagnosis tool. It does not replace a doctor or emergency care. All performance numbers will come from synthetic or public data only.

Evidence and sources: see the companion "HridayMitra ideation report" (20 pages, with links).